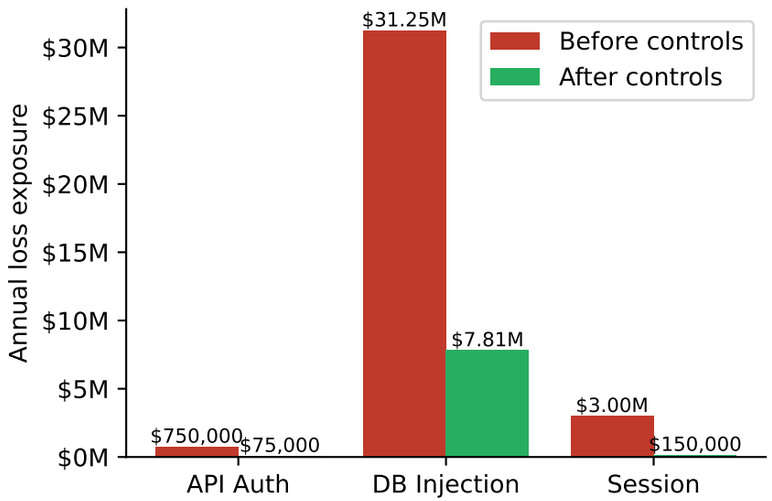
# SecureBank Financial: Cyber Risk Quantification Dashboard

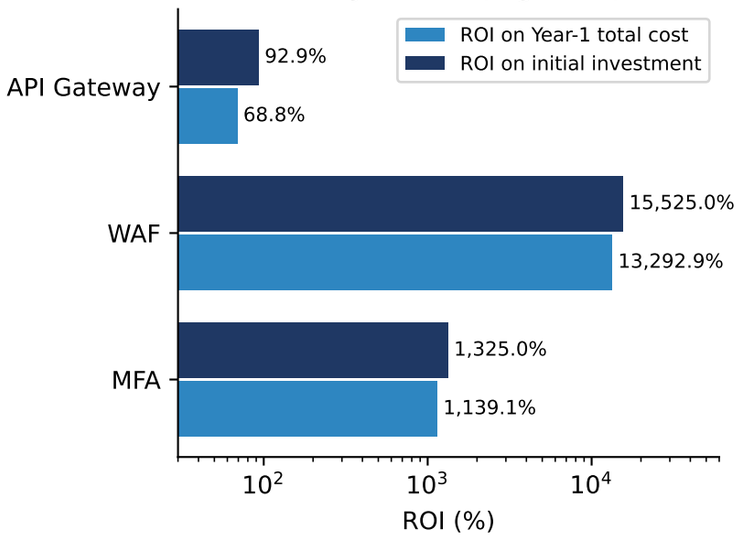
Chioma Ameh | C11/26/CGRCE/17567 | GRC102 Lab 3B | 29 Oct 2026 | Figures are annual, in US dollars

| $35.00M | $8.04M | 77.0% | $700,000 | 3,249% | 9.5 days |
|---|---|---|---|---|---|
| Total annual exposure | Exposure after controls | Risk reduction | Investment (initial) | Year-1 ROI | Portfolio payback |

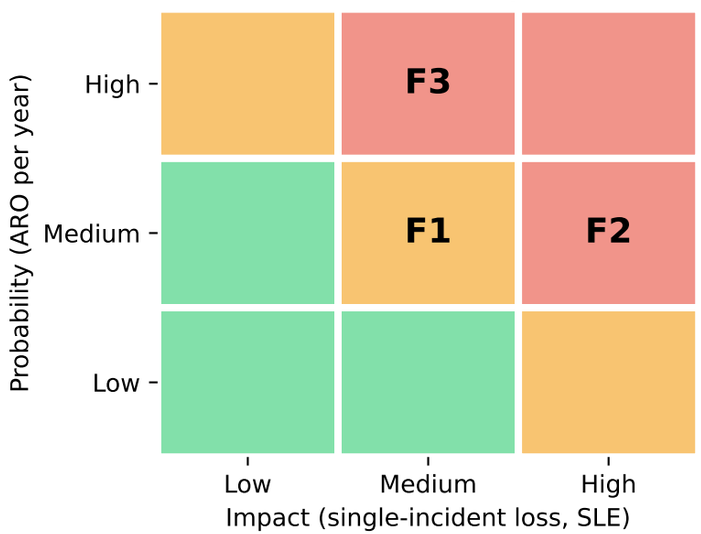
**Exposure before vs after**




**ROI by control (log scale)**




**Risk heat map**




## Top 3 risks by annual loss

| Risk | Annual loss | Priority |
|---|---|---|
| 1. Database Injection<br>Single incident $125.00M at 25% yearly probability | **$31.25M** | **High** |
| 2. Session Hijacking<br>Single incident $7.50M at 40% yearly probability | **$3.00M** | **High** |
| 3. API Authentication Bypass<br>Single incident $5.00M at 15% yearly probability | **$750,000** | **Medium** |

## Recommended controls

| Control | Cost / yr maint. | Reduces | Payback |
|---|---|---|---|
| WAF<br>Mitigates risk 1: Database Injection | $150,000 / $25,000 | **$23.44M** | 2.3 days |
| MFA<br>Mitigates risk 2: Session Hijacking | $200,000 / $30,000 | **$2.85M** | 25.9 days |
| API Gateway<br>Mitigates risk 3: API Authentication Bypass | $350,000 / $50,000 | **$675,000** | 6.7 months |

Basis: ALE = SLE x ARO; ARO = exploit probability per year (assumed). Unverified inputs: loss figures and effectiveness are as supplied in the lab sheet. See workbook for assumptions.